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7. Breadth First Search (BFS)

AIM

To write a Python program to implement Breadth First Search (BFS).

PSEUDO CODE

1. Start
2. Create graph
3. Initialize queue and visited list
4. Visit first node
5. Add adjacent unvisited nodes to queue
6. Repeat until queue is empty
7. Display traversal
8. Stop

PROGRAM

```
graph={
'A':['B','C'],
'B':['D','E'],
'C':['F'],
'D':[],'E':['F'],'F':[]
}
visited=[]
queue=['A']
while queue:
    node=queue.pop(0)
    if node not in visited:
        visited.append(node)
        queue.extend(graph[node])
print("BFS Traversal:",visited)
```

INPUT

Starting Node = A

OUTPUT

BFS Traversal: ['A', 'B', 'C', 'D', 'E', 'F']

RESULT

The BFS traversal was implemented successfully.

8. Depth First Search (DFS)

AIM

To write a Python program to implement Depth First Search (DFS).

PSEUDO CODE

9. Start
10. Create graph
11. Mark node as visited
12. Print node
13. Recursively visit unvisited adjacent nodes
14. Repeat until all nodes are visited
15. Stop

PROGRAM

```
graph={
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [], 'E': ['F'], 'F': []
}
visited=set()
def dfs(v):
    visited.add(v)
    print(v,end=' ')
    for n in graph[v]:
        if n not in visited:
            dfs(n)
dfs('A')
```

INPUT

Starting Node = A

OUTPUT

A B D E F C

RESULT

The DFS traversal was implemented successfully.